

Detected light yield governs the time resolution of RADiCAL sampling-calorimeter modules: a four-build wavelength-shifter comparison

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Abstract

We compare the precision-timing performance of four builds of the RADiCAL tungsten/LYSO:Ce sampling electromagnetic calorimeter module that differ primarily in their wavelength-shifter (WLS) capillaries – the common LYSO:Ce scintillator and tungsten absorber are unchanged – exposed to 25–150 GeV electrons at the CERN SPS: a high-light DSB1 (organic WLS) build, a low-light LuAG:Ce (ceramic WLS) build, a mixed DSB1/LuAG:Ce build, and a multi-segment build. The high-gain channels used for timing saturate, clipping at 820 mV; we recover the unclipped rising edge event by event from a linear low-gain copy of each pulse, timing a constant fraction of the low-gain-predicted true peak on an edge a factor ~ 3.7 steeper than the clipped-peak foot, which improves the resolution by a few picoseconds over timing the clipped pulse. Using a microchannel-plate-free differential estimator, $(DW - UP)/2$, formed from the four upstream and four downstream silicon-photomultiplier readouts, we measure the time resolution of every build and parametrise it as $\sigma_t = a/\sqrt{E} \oplus b$. The stochastic term a tracks the collected scintillation light, growing from 203 ± 6 to 440 ± 18 ps $\sqrt{\text{GeV}}$ — more than a factor of two — from the high-light to the low-light build, while the fitted floors of the LYSO-based builds lie in the 19–26 ps range and are mutually consistent once the capillary-count penalty of the segmented build and the extrapolation uncertainty of the low-light build are accounted for; the high-light floor, 18.8 ± 0.8 ps, confirms the 17.5 ps previously reported for DSB1 [NIM A 1068 (2024) 169737]. Within the mixed module, DSB1 and LuAG:Ce capillaries reading the same showers give per-capillary timing widths that agree at the 10–20% level ($\sigma_{\text{DSB1}}/\sigma_{\text{LuAG}} = 1.04 \pm 0.05$ under the primary estimator), which, set against the factor-of-two cross-build difference, disfavors intrinsic wavelength-shifter re-emission kinetics as the dominant cause — within this geometry and at these light levels. The high-light build reaches 25.7 ± 0.6 ps for the brightest 150 GeV showers. We discuss detected light yield as the primary design lever for fast-timing calorimetry with these modules.

Keywords: Fast timing, Sampling calorimeter, LYSO, LuAG:Ce, Wavelength shifter, Silicon photomultiplier, Signal saturation, 4D calorimetry

1. Introduction

Future high-luminosity colliders demand electromagnetic (EM) calorimeters that resolve the time of a shower to a few tens of picoseconds, in addition to its energy, to associate deposits with the correct primary vertex and reject out-of-time pile-up [1]. The RADiCAL R&D programme develops ultra-compact, radiation-hard sampling EM calorimeter modules toward this goal [2].

A first RADiCAL test-beam measurement, on a single high-light build, reported a time resolution of ≈ 27 ps at 150 GeV and parametrised the energy dependence as $\sigma_t = a/\sqrt{E} \oplus b$ with $a = 256$ ps $\sqrt{\text{GeV}}$ and a floor $b = 17.5$ ps [2]. That work also identified, as a priority for further study (its Section 7), a direct comparison of the timing performance of different wavelength-shifter (WLS)

scintillator materials in otherwise-identical modules. This paper presents that comparison.

The central question is whether a lower-light WLS material such as LuAG:Ce times worse than the high-light organic DSB1 because of the species of scintillator — its emission spectrum, decay time, or refractive index — or simply because it collects less light. The distinction matters for detector design: if light yield is the sole driver, the timing of any candidate material is predictable from its light output, and the material can be chosen for radiation hardness or mechanical properties without a separate timing penalty. We answer it by measuring four builds spanning a factor ~ 2 in light yield with a common analysis, and by separating the light-dependent stochastic term from the irreducible floor.

A prerequisite is to time all builds on an equal footing despite saturation of the front end. We therefore first describe a high-gain/low-gain (HG/LG) ratio method that recovers the saturated rising edge (Section 3), then the es-

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timator and selection (Section 4), and finally the results (Section 5): the four-build time resolution, its decomposition into a light-driven stochastic term and a material-independent floor, and the implications.

2. Experimental setup

2.1. Module and builds

The module is a tungsten/LYSO:Ce sampling calorimeter of the “shashlik” type: alternating plates of tungsten absorber and cerium-doped lutetium–yttrium oxyorthosilicate (LYSO:Ce) scintillator, ≈ 25 radiation lengths deep, read out longitudinally by WLS capillaries. Four capillaries traverse the stack and are each read out at both ends, giving $4 \times 2 = 8$ silicon-photomultiplier (SiPM) channels; we label the four downstream ends collectively DW and the four upstream ends UP. The transverse position of each incident electron is measured by a wire chamber and used to define a fiducial region on the beam spot.

Four builds were exposed, differing primarily in the WLS material of the capillaries – and hence in light yield – while sharing the same LYSO:Ce scintillator and tungsten absorber (Table 1): DSB1, high-light organic WLS; LUAG, low-light ceramic LuAG:Ce WLS; MIXED, a single module combining DSB1 and LuAG:Ce capillaries on opposite diagonals; and TENERGY, a multi-segment high-light build with an added longitudinal energy-readout section. DSB1, the brightest, is the build of Ref. [2].

2.2. Readout, beam, and reference

Each SiPM signal is split into a high-gain (HG) copy, optimised for timing, and a low-gain (LG) copy, optimised for energy linearity, both digitised by DRS4 switched-capacitor waveform digitisers (CAEN DT5742) [3] at 5 GS/s over a 1024-cell window, with per-cell calibration applied offline. The HG channels clip at ≈ 820 mV; the LG channels remain linear. Measurements used a 25–150 GeV electron beam at the CERN SPS (May 2023); DSB1 was run at all six energies in 25 GeV steps, the other builds at 50–150 GeV. A microchannel-plate photomultiplier (MCP-PMT) provides a common start; all per-channel times are MCP-referenced, and the differential estimator below cancels this reference.

3. Recovering the saturated edge: the HG/LG-ratio method

Precision timing of all builds on an equal footing requires the same well-defined edge in each, but the HG timing pulses saturate. The recorded HG amplitude piles up at the 820 mV clip (Fig. 1); for all but the smallest showers the waveform is flat across its maximum, and a constant-fraction discriminator (CFD) referenced to the measured peak is forced onto the shallow foot of the pulse, where the slope — and the timing information — is poor.

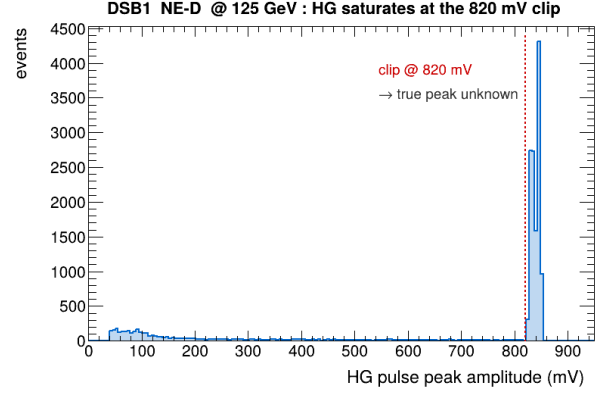


Figure 1: Distribution of recorded high-gain pulse-peak amplitudes (one channel, 125 GeV). The pile-up at ≈ 820 mV is the digitiser clip: most showers saturate, so the true peak — and the steep edge that leads to it — is not directly recorded.

The LG copy, though too small for precise timing alone, is linear and therefore measures the true amplitude the HG channel would have reached. We use a per-end calibration

$$\text{HG}_{\text{true}} = a_c + b_c \text{LG}_{\text{peak}}, \quad (1)$$

fit on the unsaturated population (Fig. 2), to predict the unclipped HG peak event by event. A CFD threshold is then placed at a fixed fraction (15%) of this predicted peak and applied to the recorded HG waveform; because the predicted peak lies far above the clip, the threshold sits on the steep, unsaturated edge. We call the resulting crossing time the saturation-recovered CFD, srCFD (analysis branch name `hg_lgcfid`); it is the primary timing estimator of this paper, with the fixed-threshold leading-edge time (LED) and the clipped-peak CFD (`cfd05`) retained as diagnostics.

The recovered edge is a factor ~ 3.7 steeper than the clipped-peak foot (Fig. 3: 381 vs 102 mV/ns). Since the slow contribution to the time resolution scales as $\sigma_t \approx \sigma_v / (dV/dt)$ with σ_v the voltage noise, a steeper edge directly improves timing. The method is applied uniformly to every build. For the low-light builds (LuAG, TENERGY), whose small pulses can defeat the fractional-threshold reconstruction, a robust fixed-threshold leading-edge time (LED) is used instead; the choice of per-build timing source is noted in Table 1.

4. Timing estimator and event selection

For each event we form the per-end mean crossing time and combine the two ends into the differential estimator

$$t_{\text{DW-UP}} \equiv \frac{1}{2}(\bar{t}_{\text{DW}} - \bar{t}_{\text{UP}}), \quad (2)$$

averaging up to four downstream and four upstream SiPM times. The difference cancels any time common to both ends — in particular the MCP start — so $t_{\text{DW-UP}}$ is free of the reference jitter; averaging four channels per end suppresses uncorrelated single-channel noise. The single-event

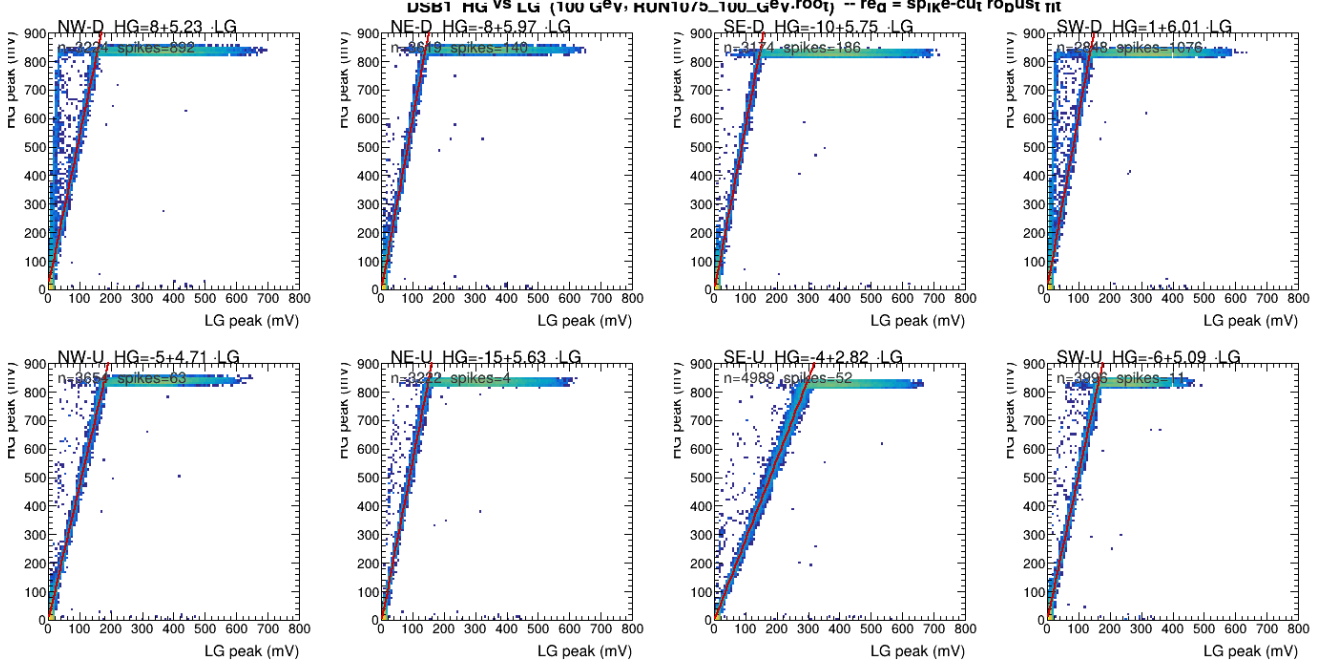


Figure 2: High-gain versus low-gain pulse-peak amplitude for the eight readout channels (100 GeV). The high-gain response saturates at 820 mV (horizontal band) while the low-gain copy remains linear; the per-channel linear fit (red line), Eq. (1), predicts the unclipped true peak.

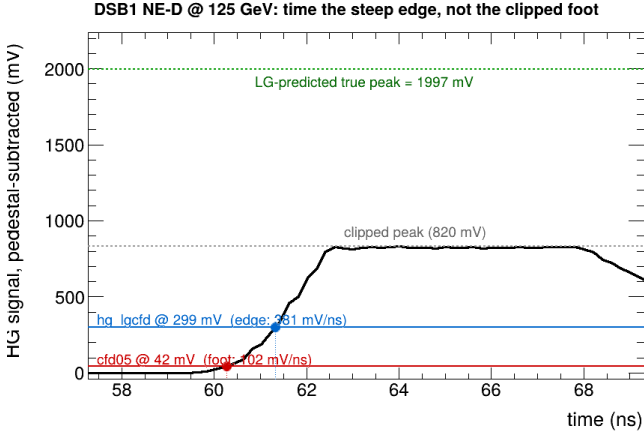


Figure 3: A single saturated high-gain waveform (125 GeV). The recorded pulse clips at 820 mV (grey); the low-gain copy predicts a true peak of 1997 mV (green). The CFD at 15% of the predicted peak (hg_lgcfid, blue) times an edge of 381 mV/ns, against 102 mV/ns at the clipped-peak foot where the 5% CFD (cfd05, red) must sit — a factor 3.7 steeper.

resolution quoted throughout is the standard deviation of a Gaussian fit to the central peak of the $t_{\text{DW+UP}}$ distribution; a small non-Gaussian shoulder (the ~ 0.2 ns satellite also noted in Ref. [2]) is excluded from the fit and has negligible effect on the quoted width.

Events are required to lie in the wire-chamber fiducial region and to have a valid MCP signal. Because shower fluctuations spread the collected light at fixed beam energy, the resolution depends on shower brightness; for a uniform, well-measured comparison across builds we quote

the resolution of the 1000 brightest showers (by total low-gain signal) at each energy, which fixes the statistical precision of every point. Within each build, the resolution improves with shower brightness, as expected from the slew argument.

5. Results

5.1. Timing distributions

Figure 4 shows, for DSB1 with all energies pooled and binned by amplitude, the $t_{\text{DW+UP}}$ (top) and absolute (DW+UP)/2 (middle) distributions with their central-peak Gaussian fits, and the fitted σ_t versus amplitude (bottom). The distributions are clean and single-peaked at every amplitude; the differential estimator is consistently narrower than the absolute one, as expected once the common-mode reference jitter is removed, and the width falls monotonically with shower brightness.

5.2. Four-build timing response

Figure 5 is the central energy-scan result. For each build, the brightest-shower σ_t is plotted against beam energy and fit to the published form

$$\sigma_t(E) = \frac{a}{\sqrt{E}} \oplus b, \quad (3)$$

with a the (light-driven) stochastic term and b the floor as $E \rightarrow \infty$. The fitted parameters are collected in Table 1. Two features stand out. First, the builds order by light yield, and the stochastic term tracks it: a rises from

DSB1 (lgcfd): per-amplitude-bin (DW-UP)/2 [top] and (DW+UP)/2 [middle] distributions, with their fitted σ_t [bottom]

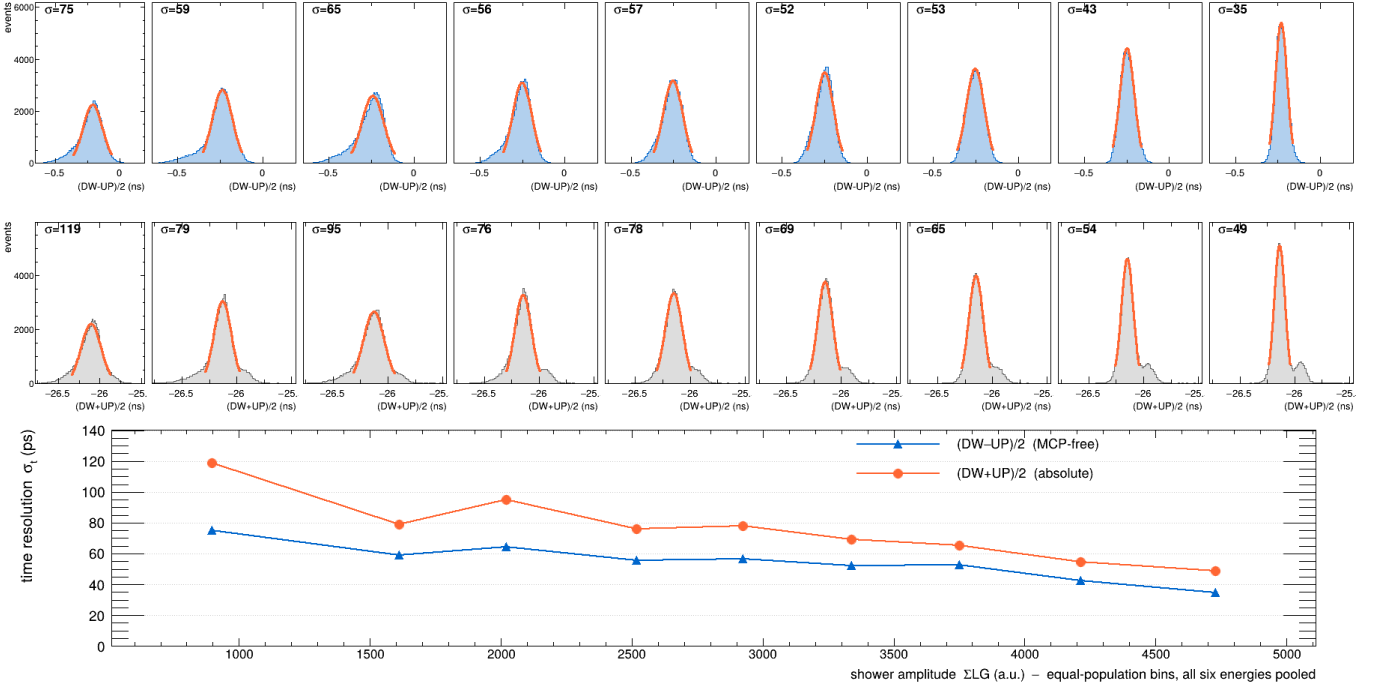


Figure 4: DSB1 timing distributions, all energies pooled and binned by amplitude. Top: $(DW - UP)/2$ (MCP-free). Middle: $(DW + UP)/2$ (absolute). Each σ (ps) is the central-peak Gaussian fit. Bottom: the fitted σ_t versus amplitude. The peaks are clean Gaussians at every brightness.

203 ± 6 ps $\sqrt{\text{GeV}}$ for high-light DSB1 to 440 ± 18 ps $\sqrt{\text{GeV}}$ for low-light LuAG — more than a factor of two. Second, the floors of the LYSO-based builds lie in the 19–26 ps range: 18.8 ± 0.8 (DSB1), 26.0 ± 1.8 (TENERGY), 24.6 ± 3.3 ps (LuAG). These are mutually consistent once two effects are accounted for: TENERGY times with three capillaries rather than four (a $\sqrt{4/3} \approx 1.15$ statistical penalty, bringing its four-capillary-equivalent floor to ≈ 22 ps), and the LuAG floor is an extrapolation — its best measured point is 44.4 ps at 150 GeV. The DSB1 floor confirms the 17.5 ps reported in Ref. [2]. (MIXED sits higher, $b = 34.3 \pm 2.4$ ps, for a methodological reason: a single module-wide timing estimator is ill-posed for a module in which half the capillary ends saturate and half do not, so its row in Table 1 is a reference value only; the physically meaningful MIXED comparison is the same-shower analysis of Sec. 6.) The energy dependence is well described by the $1/\sqrt{E}$ (photon statistics) form; a $1/E$ (pure-slew) form fits markedly worse.

The interpretation: detected light yield governs the dominant, material-dependent (stochastic) part of the time resolution, while the fitted floors are consistent within their uncertainties and caveats. A lower-light configuration times worse in proportion to its reduced light. Whether the residual cross-build difference reflects the wavelength-shifter species (its re-emission kinetics) or only its light output cannot, however, be decided from the energy scan alone — a between-build comparison cannot sep-

Table 1: Per-build fit of $\sigma_t = a/\sqrt{E} \oplus b$ to the brightest-shower resolution, the timing source used, and the resolution measured at 150 GeV (statistical errors; selection systematics in Table 2). The WLS column gives the wavelength-shifter material of the capillaries; the LYSO:Ce scintillator and tungsten absorber are common to every build. The published DSB1 values (cfd05) are shown for comparison. TENERGY carries a $\sqrt{4/3}$ capillary-count penalty; the MIXED module-wide row is a reference value (single-estimator timing is ill-posed there, Sec. 6).

Build	WLS	src	a (ps $\sqrt{\text{GeV}}$)	b (ps)	σ_t^{150}
DSB1	organic, hi	srCFD	203 ± 6	18.8 ± 0.8	25.7
TENERGY	organic, seg	LED	198 ± 21	26.0 ± 1.8	30.0
MIXED	org.+cer.	srCFD	253 ± 29	34.3 ± 2.4	39.0
LUAG	ceramic, lo	LED	440 ± 18	24.6 ± 3.3	44.4
DSB1 [2]	organic	cfd05	256	17.5	20.0

arate the two. The mixed module provides the controlled, same-shower test (Sec. 6).

5.3. The gain from recovering the edge

The benefit of the HG/LG-ratio method is isolated by a direct comparison on identical showers (Fig. 6): the brightest-1000 DSB1 resolution is computed with cfd05 (a CFD on the clipped measured peak, as in Ref. [2]) and with srCFD (the recovered edge; hg_lgcfd), the selection and estimator held fixed so that only the per-channel time differs. Timing the recovered edge improves

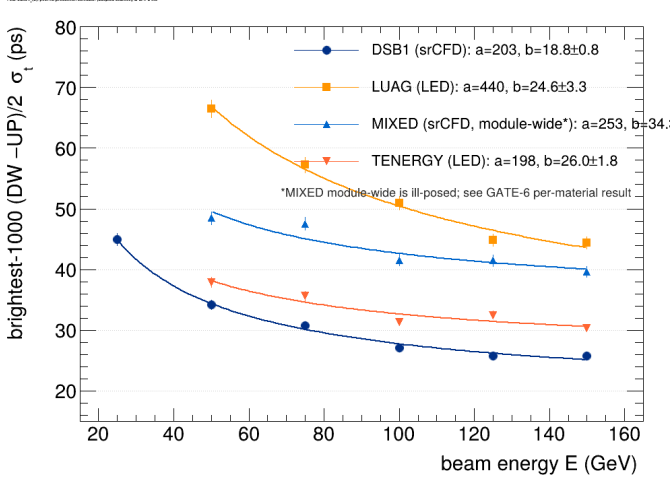


Figure 5: Brightest-shower (DW - UP)/2 time resolution versus beam energy for the four builds, each fit to $\sigma_t = a/\sqrt{E} \oplus b$ [Eq. (3)], using the production estimator chain. The stochastic term a tracks light yield (DSB1 lowest, LuAG highest). The MIXED module-wide curve is a reference only (see text and Sec. 6).

the resolution by a few picoseconds at 150 GeV and lowers the fitted floor. The gain is energy-dependent, largest where the timing is hardest: at high energy, where saturation is most severe, and at the lowest energy, where the 5% foot of a small pulse sits in noise; it is small at 50–75 GeV. The stochastic term is essentially unchanged, so the method acts chiefly on the high-energy/floor regime. (The cfd05 stochastic term in this controlled comparison reflects the brightest-shower selection and differs from the 256 ps $\sqrt{\text{GeV}}$ of Ref. [2] in Table 1, which used a different selection; holding the selection fixed isolates the timing method.) The best single-event resolution measured, for the brightest 150 GeV DSB1 showers, is 25.7 ± 0.6 ps.

5.4. Systematic uncertainties

The headline brightest-1000 (DW - UP)/2 resolution was recomputed under deliberate variations of every selection knob, on the adopted timing source: the brightness selection ($K = 500$ and 2000 versus the nominal 1000), the timing fiducial radius (2.5 and 3.5 versus 3.0 mm), the MCP saturation window (upper cut 700 versus 750 mV), the per-channel HG threshold (30 versus 20 mV), and the fit energy range (50–150 versus 25–150 GeV). The shifts are collected in Table 2, with the total systematic taken as their RMS. The selection values themselves were fixed by the scans of Appendix Appendix A (Fig. A.9); the timing-method choice (cfd05 versus the recovered edge) is the subject of Sec. 5.3 and is not folded into this budget.

The headline is robust: the total selection systematic on $\sigma_t(150)$ is ≤ 1 ps for the high-light builds (DSB1, TENERGY) and ~ 1.5 ps for MIXED and LUAG. The dominant single variation is loosening the brightness selection to $K = 2000$, which admits dimmer showers and genuinely degrades the resolution by 2–2.4 ps — a manifestation of

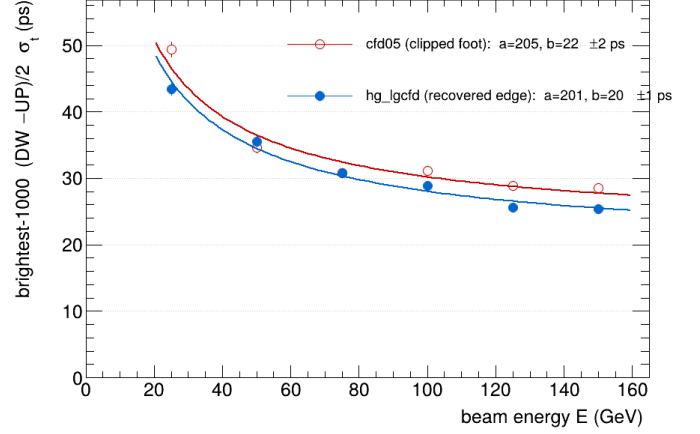


Figure 6: Time resolution of the brightest-1000 DSB1 showers timed with cfd05 (CFD on the clipped foot, open) and hg_lgcfd (CFD on the LG-recovered edge, filled), identical events and selection. Recovering the edge improves the resolution by ~ 3 ps at 150 GeV and lowers the floor.

the same light-resolution dependence that is the paper’s subject, not an analysis artefact. Crucially, the DSB1 floor is stable under these variations, so its agreement with the published 17.5 ps survives the systematic budget. The elevated floor of the MIXED module-wide fit is the ill-posed-estimator feature discussed in Secs. 5.2 and 6 and carries the largest systematic.

6. The same-shower control: DSB1 versus LuAG:Ce in the mixed module

The energy scan of Sec. 5.2 cannot by itself distinguish whether the low-light build times worse because of its wavelength-shifter species (slower re-emission kinetics) or because of its reduced light. The MIXED module breaks this degeneracy: two of its four capillaries carry DSB1 and two carry LuAG:Ce, on opposite diagonals, so both materials read the same showers in the same LYSO:Ce scintillator, the same module, and the same digitiser group. The corner-material assignment is confirmed by amplitude-independent pulse-shape discriminants (low-gain charge-to-peak ratio and constant-fraction rise time) validated on the single-material modules; all eight capillary ends classify consistently.

For each event with all timing ends valid, we form the in-event difference of the two same-material downstream ends — $t_{NE} - t_{SW}$ for DSB1 and $t_{NW} - t_{SE}$ for LuAG:Ce — whose widths measure the per-capillary timing jitter of each material with beam arrival time, group timebase, and the MCP reference cancelling by construction. Figure 7 shows the result under the primary srCFD estimator. The per-capillary widths of the two materials agree at the 10–20% level at every energy; the width ratio $\sigma_{\text{DSB1}}/\sigma_{\text{LuAG}}$ is 0.94, 1.06, 0.93, 1.07, and 1.19 at 50–150 GeV, with a five-energy mean of 1.04 ± 0.05 , where the uncertainty is the energy-to-energy scatter (which dominates the per-point statistical

Table 2: Systematic budget for the brightest-1000 (DW – UP)/2 time resolution. Each row recomputes $\sigma_t(150)$ and the fit floor b under one selection variation; the entry is the shift from the nominal value (ps). The total systematic is the RMS of the shifts; the statistical floor error is from the fit. The timing-method choice (cfd05 vs. the recovered edge) is the subject of Sec. 5.3 and is not folded in here.

Variation	DSB1	TENERGY	MIXED	LUAG
shift in $\sigma_t(150)$ [ps]				
$K=500$	+0.1	-0.3	-1.0	+0.8
$K=2000$	+2.1	+2.4	+2.3	+0.4
$r_{\text{fid}}=2.5$ mm	+1.2	+0.3	-0.7	-1.1
$r_{\text{fid}}=3.5$ mm	+0.2	+0.6	+2.0	+3.8
MCP < 700 mV	+0.3	+0.2	+2.5	-0.7
HG > 30 mV	+0.0	-0.0	+0.0	+0.0
fit 50–150 only	+0.0	+0.0	+0.0	+0.0
nominal $\sigma_t(150)$	25.3	28.3	39.1	41.9
total syst.	± 0.9	± 0.9	± 1.5	± 1.6
floor b (nom)	19.5	21.6	34.6	19.8
b syst.	± 1.2	± 2.2	± 2.2	± 2.4
b stat.	± 1.1	± 2.7	± 2.4	± 5.6

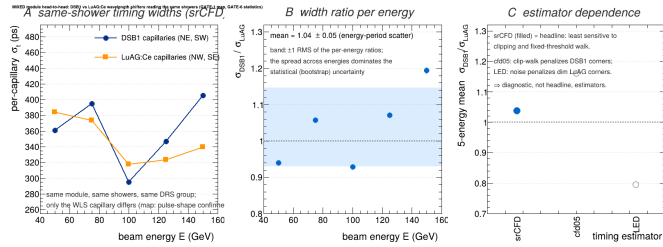


Figure 7: Same-shower comparison in the MIXED module (corner map confirmed by pulse-shape discriminants). Left: per-capillary timing widths versus energy for the DSB1 (NE, SW) and LuAG:Ce (NW, SE) capillary pairs under srCFD; widths are same-layer invariant pair differences divided by $\sqrt{2}$, so common-mode terms cancel. Centre: the width ratio per energy with the ± 1 RMS energy-scatter band; the scatter dominates the statistical uncertainty. Right: estimator dependence (srCFD headline; cfd05 and LED diagnostic).

uncertainty). The ratio is estimator-dependent at a comparable level — 1.04 (srCFD), 1.16 (cfd05, whose clip-walk penalises the saturating DSB1 corners), 0.80 (LED, whose fixed threshold penalises the dimmer LuAG corners) — so srCFD, the least clipping- and walk-sensitive choice, defines the headline, and the others are diagnostic. Two caveats apply: the pair widths contain a common transverse position-coupling term that dilutes the ratio toward unity, and the comparison is specific to this geometry and these light levels.

Set against the more-than-factor-of-two stochastic-term difference between the pure builds, this same-shower near-parity disfavors intrinsic wavelength-shifter re-emission kinetics as the dominant cause of the cross-build timing difference. It does not establish universal material equivalence: at lower light levels or in other geometries the kinetics could re-emerge.

The depth dial: shower-max migration read as a dual-end propagation-time shift, $\Delta(t_{\text{DW}} - t_{\text{UP}})/2$ (E)

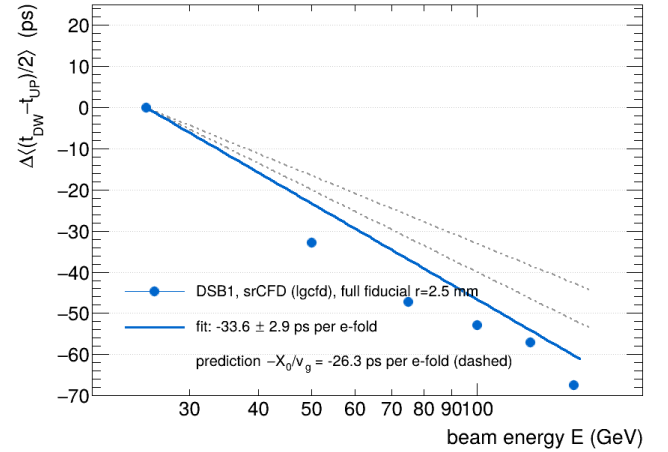


Figure 8: Energy dependence of the mean dual-end timing asymmetry $\Delta((t_{\text{DW}} - t_{\text{UP}})/2)$ (DSB1, srCFD, full fiducial), relative to 25 GeV. The dashed band is the $-X_0/v_g$ expectation from shower-maximum migration. The fitted slope is -33.6 ± 2.9 ps per e -fold (acceptance-conditional; see text).

7. Longitudinal timing asymmetry: the depth dial

The (DW – UP)/2 estimator is, by construction, sensitive to the longitudinal position of the light emission: a deeper shower is closer to the downstream ends, shifting the asymmetry negative. Shower maximum migrates downstream as $\sim X_0 \ln E$, predicting a drift of the mean asymmetry of $-X_0/v_{\text{eff}} \approx -26$ ps per e -fold of energy for axial light propagation. The measured drift, on the full fiducial sample with the amplitude-independent srCFD estimator, is monotonic and negative with slope -33.6 ± 2.9 ps per e -fold — consistent in sign and magnitude with the shower-maximum migration (the $\sim 30\%$ excess over the axial estimate is consistent with modal dispersion lowering the effective propagation speed). The drift is read only by constant-fraction estimators; fixed-threshold times, which fire on the earliest photons, suppress it. The slope is an acceptance-weighted population quantity (the fixed-position wavelength-shifter filament samples different shower depths with different efficiency), so we use it here only as a consistency reading: the event-to-event spread of the same asymmetry is the timing floor of Sec. 5.2, and its mean drift confirms that the floor carries genuine longitudinal shower information rather than instrumental jitter alone.

8. Discussion

These measurements establish detected light yield as the primary design lever for the timing of RADiCAL modules: the cross-build stochastic term tracks the light, and the same-shower control of Sec. 6 disfavors wavelength-shifter kinetics as the dominant alternative explanation, within this geometry and at these light levels. On this evidence a WLS material may be selected for radiation hardness,

mechanical robustness, or cost, with its timing then estimated from its light output. LuAG:Ce, attractive for its radiation hardness [4], times worse than DSB1 at a level commensurate with its lower light yield; the same-shower comparison bounds any additional species-specific penalty at the 10–20% level under the primary estimator.

The fitted floors of the LYSO builds lie in the 19–26 ps range, consistent within their uncertainties and the stated caveats, and the DSB1 floor confirms the previously published value. The floor’s origin is plausibly the shower itself: the $(DW - UP)/2$ estimator is sensitive to the longitudinal position of energy deposition, and the measured energy drift of its mean (Sec. 7) is consistent with the shower-maximum migration — so shower-to-shower depth fluctuation produces an irreducible spread that does not decrease with light and is common to any scintillator filling the same geometry. This same depth sensitivity suggests the route below the floor: a depth-corrected estimator, using the dual-end information to estimate and subtract each shower’s depth, which we are pursuing.

9. Conclusions

Comparing four WLS builds of a tungsten/LYSO:Ce sampling calorimeter on a common footing — enabled by a high-gain/low-gain-ratio method that recovers the saturated rising edge — we find that the time resolution is governed primarily by detected scintillation light yield and module response rather than by wavelength-shifter identity alone. The stochastic term of $\sigma_t = a/\sqrt{E} \oplus b$ grows from 203 ± 6 to 440 ± 18 ps $\sqrt{\text{GeV}}$ — more than a factor of two — from the high-light to the low-light build, while in the mixed module the two wavelength-shifter species reading the same showers give per-capillary timing widths agreeing at the 10–20% level (1.04 ± 0.05 under the primary estimator). This combination disfavors intrinsic wavelength-shifter re-emission kinetics as the dominant explanation for the cross-build difference, within this geometry and at these light levels. The fitted floors of the LYSO builds lie in the 19–26 ps range, and the high-light floor of 18.8 ± 0.8 ps confirms the previously published 17.5 ps. The high-light build reaches 25.7 ± 0.6 ps for the brightest 150 GeV showers, and the mean dual-end asymmetry drifts with energy in quantitative agreement with shower-maximum migration, identifying the floor with longitudinal shower physics. Detected light yield is therefore the design lever for fast-timing calorimetry with these modules, and the depth-corrected estimator it suggests is the target of the analysis to follow.

Appendix A. Selection optimization and systematic stability

The four choices that define the headline estimator were each fixed by a scan at the headline energy (150 GeV), shown for all builds in Fig. A.9. Panel (a) compares timing sources: a constant-fraction time on the clipped peak,

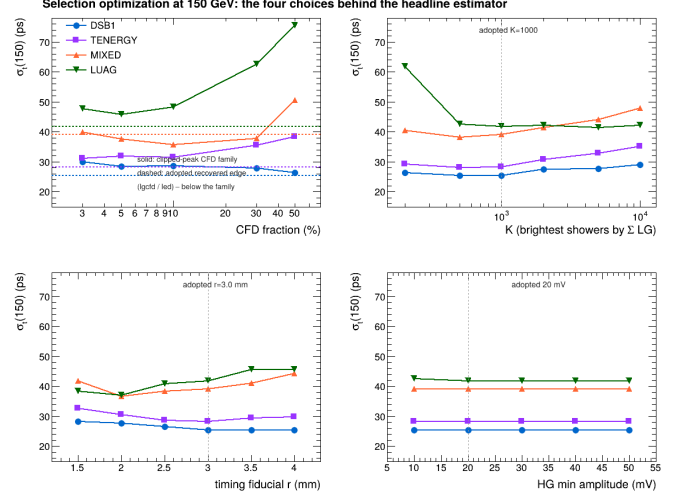


Figure A.9: Selection optimization at 150 GeV for all four builds: (a) timing source as a function of CFD fraction on the clipped peak (solid) with the adopted recovered-edge source overlaid (dashed); (b) resolution versus the brightest- N selection K ; (c) versus timing fiducial radius; (d) versus the per-channel HG threshold. Dashed grey lines mark the adopted working points.

swept over fractions 3–50% (solid), against the recovered-edge source we adopt (srCFD/LED, dashed). The adopted source lies below the entire clipped-peak family for every build — dramatically so for the low-light LUAG — which is the quantitative basis for the method of Sec. 3. Panel (b) shows the brightness selection: σ_t falls steeply as the dimmest showers are removed and flattens near $K \sim 1000$, beyond which admitting dimmer showers degrades it again; $K = 1000$ sits at the knee. Panel (c) shows the resolution is flat across the timing fiducial radius out to the chosen 3.0 mm, rising only as off-axis showers leak in. Panel (d) shows insensitivity to the per-channel HG threshold, confirming the 20 mV working point is well clear of the noise floor. The corresponding systematic stability of $\sigma_t(150)$ under these variations is summarised in Fig. A.10 and Table 2.

Acknowledgements

We thank the CERN SPS staff for the test-beam operation.

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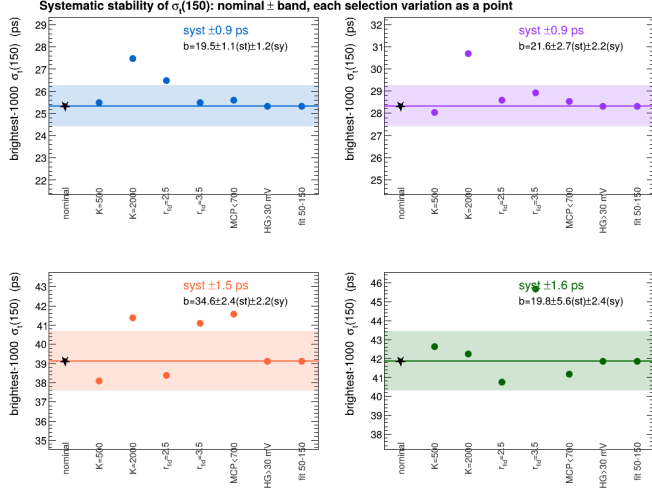


Figure A.10: Systematic stability of the brightest-1000 $\sigma_t(150)$ for each build: the nominal value (star) with its total-systematic band, and each selection variation as a point. The fitted floor b with its statistical and systematic errors is annotated per build. All variations lie within the band except the deliberate $K = 2000$ loosening of the brightness selection.

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